Loop Gain — translation thesis

Draft, 2026-10-28, for Jeff to rewrite in his words and Lex to review against the four questions. Revised after Lex's speaker-session notes on issue 10: one carried relationship, the disclosure split four ways. Word count is kept under 500 for the body. Drafted by Claude, working with Jeff.

1. The source. A finding from a week of our own research on two models. In a feed, every frame the reader applies re-enters as signal, and the echo does not act. In the exclusion loop of Lex's second brief, a signal pushed out of awareness returns, invites the act that produced it, and comes back louder. Turn the drive up: does the end state slide or drop? The feed slid. The loop dropped, where its reproduction number predicted. One loop at two gains; the difference is one rule, whether what returns invites an act. Chladni's plate, 1787, is the oldest instance of what lies past the threshold: the sand reorganises into a figure rather than sliding.

2. What we carried across, and in which layer. The structure of feedback reproduction, at the finding layer, not feeds or symptoms. One stream of pings is read by three grammars of return. In the first a gesture comes back once, as itself: a slope. In the second the same pings pass through a delay with gain, so a return is excluded again and returns louder; the gain is the reproduction number, and when it crosses one the echoes stop dying and bloom into a howl. In the third the howl's settled note drives the plate, and the sand gathers into its figure. Return, reproduction, gain, threshold, form.

3. What changed. Models, benches and proofs became sound, delay and gain, a visible level curve, a transition at reproduction number one, and a pattern on a plate. We kept the threshold and the plate's mathematics. We simplified the feed to a headless run of its own model with its sound. We distorted the room: the loop's resonance is a filter at 870 Hz placed by hand so the howl settles where the plate has a figure, and a soft clip stands where a speaker's ceiling would be. At Lex's filter we added Bridson's growth as the exterior and Atlas as the piece's map. All of it is written on screen. We left out the body, Lex's allostatic load, which the bench carries in four frames and three minutes cannot hold. The movements are cut by the clock, not by the sound; a truer version would wait for the howl.

4. What making it revealed. What the reproduction number does not say. A fourth frame, from Wegner's ironic process theory, had load raise the return rate. We predicted it would barely move the threshold, and it did not; we predicted the frame would be small, and were wrong by an order of magnitude. Doubling the return rate halves the time a generation takes. R0 says whether; the return rate says how fast. We proved the first half as theorems and could only watch the second; a three-minute window hid the divergence our first test was built to find. The piece is about that division, between what a ratio tells you and what needs a clock, and we did not know it when we started.

Source citation. Highman and Krížová, *The Unframeable Now* and *The Return of the Excluded*, concept briefs, benches and proofs, in this repository, October 2026. E. F. F. Chladni, *Entdeckungen über die Theorie des Klanges* (Leipzig, 1787); the plate model follows Hans Jenny, *Kymatik* (1967), for the round plate's Bessel modes and crossed standing waves for the square. The loop's threshold is the branching-process criterion, reproduction number one; its sound is the Larsen effect. D. M. Wegner, "Ironic processes of mental control", *Psychological Review* 101 (1994).

AI-use disclosure. Claude (Anthropic) worked with Jeff Highman throughout. Research and brainstorming: Claude surveyed the literature behind the fourth frame and proposed the slope-or-cliff question and the reproduction-number reading. Coding: Claude wrote the feed model, the exclusion bench, the plate port, the piece and the film under Jeff's direction. Mathematical and editorial support: Claude wrote the Lean proofs and the drafts of this thesis, the process note and the paper. Human work: Jeff chose the source, the constants, the three movements, the resonance and what to leave out, and made every decision about what to submit; Alexandra Krížová wrote the two briefs, extended the second with the allostatic reading, set the translation test every result was read against, and reviewed with a different AI assistant so the two would not share blind spots. Alexandra Krížová — critical reviewer and development reviewer.

The take. `loop-gain.mp4`, recorded from the page on [date], [length].